# Machine – Calibrate

**Menu: Machine > Calibrate**

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

## Step-by-Step Calibration

### ① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

### ② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is displayed under the lock (for motor controllers).
- RC stops automatically when the target rate is met.

*If RC fails to reach target RPM:*

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

### ③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

### ④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.

## Controls

| Control | Action |
|---|---|
| Start | Start the meter roller / begin collecting a sample |
| Stop | Stop the meter roller |
| Lock/Unlock | Lock or unlock the current RPM setting |
| Power on/off | Switch the product on or off |
| Cancel | Discard changes |
| Save | Save the calculated Cal Factor |

## Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.
- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.